

Lecture 5

USENET

Overview

- ▶ USENET Introduction and Theory
- ▶ History of USENET
- ▶ USENET Structure and Operation
- ▶ USENET Issues
- ▶ Summary

Why Should You Care?

- ▶ USENET News is typically provided as a matter of course by Internet service providers
 - ▶ A check-box item
- ▶ USENET “push” model of content transmission is still useful
 - ▶ as the proliferation of “groupware” would demonstrate
- ▶ USENET can be very resource intensive
 - ▶ Bandwidth, hardware, management personnel
- ▶ USENET articles can get you into trouble

USENET Introduction and Theory

- ▶ USENET is
 - ▶ A content transport system
 - ▶ Like electronic mail, only different
 - ▶ A logical network layered on top of other networks
 - ▶ A broadcast, one-to-many medium
- ▶ Derived from very early Unix networking technology
 - ▶ “Unix to Unix Copy (UUCP)”
- ▶ Internet USENET hosts normally runs its own protocol (NNTP) over TCP/IP, but can also use UUCP over TCP
 - ▶ UUCP over TCP useful in **very** bad network conditions

USENET Structure and Operation

- ▶ A distributed Bulletin Board System
 - ▶ Take your message and “post” it on the BBS
- ▶ Users post messages (“articles”) to areas called “newsgroups”
 - ▶ Newsgroups have themes or topics
 - ▶ Each article is given
 - ▶ a site relative article number
 - ▶ a globally unique message identifier
- ▶ Articles can be posted to multiple newsgroups at one time
 - ▶ Frowned upon, but common
- ▶ Articles are copied to other USENET news servers
 - ▶ All servers willing to accept the article on the entire Internet

USENET Structure & Operation (cont'd)

- ▶ No central control or authority
 - ▶ Anyone can create and post a news article
- ▶ New newsgroups can be created by anyone
 - ▶ simply post a specially formatted article called a “control message”
 - ▶ Control messages are easily forged
 - ▶ Can be cryptographically signed using PGP
- ▶ Local policy determines how long articles are kept in storage

Newsgroups

- ▶ Newsgroup hierarchies vary wildly
- ▶ Acceptance of a particular hierarchy is a local policy decision

Newsgroups (cont'd)

- ▶ Newsgroups can be moderated or unmoderated
 - ▶ Articles posted to a moderated newsgroup must have an approved header field
 - ▶ The moderator is supposed to be the one to do this
 - ▶ Easy to forge the appropriate magic to get past this check
 - ▶ Moderators are volunteers interested in the subject of the newsgroup
 - ▶ Most newsgroups are not moderated
 - ▶ However, moderated ones usually have better signal to noise ratios

Flood Fill Article Propagation

- ▶ USENET articles are propagated using “flood fill”
 - ▶ Each USENET news server has one or more peers
 - ▶ Each article received from a peer or from a user of that server (a posting) is sent to all other peers that haven’t yet seen the article
 - ▶ A “push” model of data transmission
- ▶ The full article is copied all over the Internet
 - ▶ when using real-time feeders, the article can reach all major news hosts on the Internet in a matter of minutes

Article Format

- ▶ Plain 7-bit ASCII text
 - ▶ non-ASCII encoded into ASCII
 - ▶ typically using *uuencode*
 - ▶ MIME encoding becoming more and more popular
- ▶ Resembles an email message
 - ▶ News article format a subset of e-mail (RFC 822) format
- ▶ Described in RFC 1036
 - ▶ “Son of RFC 1036” is in progress
- ▶ Content of articles moving more and more to HTML
- ▶ Has 6 required headers
 - ▶ `From` -- who wrote the article
 - ▶ `Date` -- the date the article was posted
 - ▶ `Newsgroups` -- the newsgroup(s) the article was posted to
 - ▶ `Subject` -- the subject of the article
 - ▶ `Message-ID` -- a globally unique identifier for the article
 - ▶ `Path` -- the UUCP path the article has taken to reach the current system
- ▶ Other headers optional
 - ▶ Unknown headers passed unchanged

NNTP and Its Use

- ▶ NNTP is a simple application layer protocol
 - ▶ “Standard” verb/numeric response code format
 - ▶ Described in RFC 977
- ▶ Mostly a command/response protocol
 - ▶ One server sends “I have article <number>” to peer
 - ▶ Peer sends “no thanks, seen it already” or “OK, send it”

Internet Network News

- ▶ ISC's INN is an open source USENET news system
 - ▶ Available from <ftp://ftp.isc.org/isc/inn/inn.tar.gz>
- ▶ INN is a transport system
 - ▶ Will use an appropriate application layer transport mechanism
 - ▶ NNTP (by preference)
 - ▶ UUCP
 - ▶ even SMTP
 - ▶ Can also handle compressed batches of news
 - ▶ Can be extended easily to handle other transport mechanisms as needed

What INN Does

- ▶ Transport news articles
- ▶ Implements NNTP (RFC 977)
- ▶ Primarily uses TCP/IP
 - ▶ Can use UUCP or other transport mechanisms
- ▶ Provides network client (reader) interface
- ▶ Feeds in real-time or in batch mode
 - ▶ Compressed or uncompressed articles

What INN Doesn't Do

- ▶ No client software (news readers)
- ▶ No extra support for large-scale “reverse” (sucking) feeds
 - ▶ “Pull” model instead of “push”
- ▶ No web interfaces for users or administrators (yet)
 - ▶ Management of INN is a painful
- ▶ INN is middleware and not a vertical solution
 - ▶ Vertical solutions such as Netscape's Collabra exist

Types of News Servers

▶ Transit servers

- ▶ Usually at enterprise gateways
- ▶ Have no regular reader clients
- ▶ Don't keep articles around for long
- ▶ Less resource requirements than readers
- ▶ Easier to secure

▶ Reader Servers

- ▶ Require significantly more resources than transit servers
- ▶ Require more management resources
- ▶ Usually stores articles for long periods
- ▶ Targets for spammers

Caching NNTP Servers

- ▶ Provides some level of scalability
 - ▶ Reduced resource requirements, higher performance
- ▶ When a reader requests articles, the caching server first checks local storage and (if article isn't found) requests the article from an upstream server using NNTP reader commands
 - ▶ Upstream server treats the request like any other reader request
- ▶ Articles typically fetched on demand, but large numbers of articles can be pre-fetched

Caching NNTP Servers (cont'd)

- ▶ Lets the site running the caching server avoid accepting a full feed
 - ▶ Full feeds demand large amounts of disk space
- ▶ Useful for sites with inconsistent or sparse reading patterns
- ▶ Not a good idea for sites with poor network connections
 - ▶ Reader performance affected by upstream server

Futures

- ▶ “Groupware” such as Lotus Notes and Netscape Collabra are the next evolutionary of USENET News
 - ▶ Very pretty user interfaces on the news reading clients
 - ▶ Much more easily managed servers
 - ▶ Tighter integration of transport / user interface / article store
 - ▶ Includes database retrieval mechanisms for article content
- ▶ USENET messages will likely become more HTML rich
 - ▶ Newsreaders unable to handle HTML will likely fade away
- ▶ USENET will continue to evolve

USENET Evolution

- ▶ Current USENET technology results in tremendous resource utilization
 - ▶ Disk, network, CPU, management, etc.
- ▶ Gigabytes / day of messages
 - ▶ Typically, only a tiny percentage of these messages are ever read
 - ▶ Large percentage of messages are spam

USENET Evolution (Cont'd)

- ▶ USENET articles will likely move to a header/pointer format
 - ▶ Content only fetched if article is read
 - ▶ Gateways to “old” USENET that fetch the content and create/post a “legacy” article
- ▶ Likely permits a reduction in the amount of resources consumed
 - ▶ Can be aided by integration with WWW caches
- ▶ Can help in the reduction of spam
 - ▶ Integration with tools like MAPS/RBL

USENET Issues

- ▶ As with any service which provides content, “inappropriate” content can be found
 - ▶ Hateful literature, pornography, libel/slander
 - ▶ There are constant calls to censor this content
 - ▶ ISPs often get caught in the middle
 - ▶ Easy targets
 - ▶ Little control
 - ▶ Technological advances may “help” content control issues
- ▶ USENET growth will continue to be an issue
 - ▶ New technology may help this as well

Summary

- ▶ USENET has been around since the beginning of the Internet
- ▶ News is still useful for pushing information to a wide audience
 - ▶ A flood fill model of information propagation assures global distribution
- ▶ USENET News hierarchy is largely chaotic
- ▶ USENET articles are similar to mail format
- ▶ USENET will likely evolve to a header/pointer format
 - ▶ Will reduce the resource requirements and (hopefully) help the signal to noise ratio



Questions ?